

WMDR2

Package in package 'Model'

WMDR2

Version 0.3.3 Phase 1.0 Proposed

jdk created on 06.03.2026. Last modified 21.09.2026

WMDR2 diagram

Class diagram in package 'WMDR2'

WMDR2 v0.3.x

v0.3.0

- Cardinality of facility.deployments changed to [0..*]
- Deployment.location [0..1] added to describe relative location of instrument as free text.
- Deployment.serialNumber moved to ObservingConfiguration.InstrumentInstance.serialNumber
- Deployment.instrument associated to InstrumentInstance
- Deployment.sourceOfObservation moved to ObservingConfiguration
- As a consequence of the above, Deployment describes the location context only and is renamed to ObservingLocation.
- "reporting" folded into "ReportingProcedure"
- ReportingProcedure.validFrom dropped.
- Reporting.spatialReportingInterval added.
- cardinality contact.organization from [0..1] to 1
- cardinality wmoRegion from [0..1] to 1
- cardinality of ReportingProcedure.dataPolicy from [0..1] to 1; nilReason allowed
- cardinality of ReportingProcedure.internationalExchange from [0..1] to 1
- cardinality of ObservingConfiguration.operatingStatus from [0..1] to 1
- cardinality of ObservingConfiguration.sourceOfObservation from [0..1] to 1

v0.3.1

- remove ObservingLocation, move all attributes to ObservingConfiguration.
- rename validFrom to time, which is specified as an OGC period / interval.
- move serialNumber directly under observingConfiguration, drop separate instrumentInstance.
- observingConfiguration.operatingStatus changed from 1 to 0..1

v0.3.2

- remove association linking facility with ObservingConfiguration as it is not needed.
- v0.3.3 alignment with github.com/wmo-im/wmdr2
- rename ObservationSeries to Observation and the associations from observationSeries to observations.
- rename ObservingConfiguration to Configuration and the associations from observingConfigurations to configurations.
- rename Configuration.serialNumber to Configuration.instrumentSerialNumber.
- rename instrument to Instrument, correct cardinality.
- remove Configuration.verticalDistanceFromReferenceSurface
- remove Configuration.referenceSurface
- add class VerticalDistance as Configuration.verticalDistance[0..1] with distances[1..*], unit, referenceSurface.
- add Configuration.id
- update association of Observation with ProgramAffiliation[1..*]
- update Facility.territories to 0..* as this may not apply for all facilities.
- Rename temporalGeometry.time[1..*] to .date[1..*]. These are individual dates aligned with coordinates, not one OGC time object. Rename association from temporalGeometries to temporalGeometry.
- make Contact genuinely reusable in different contexts; add note that in the WMDR2 JSON representation, a contact can have different roles depending on the context.
- Make Schedule genuinely reusable, to be implemented as JSCalendar object. Move WMO extensions to ReportingProcedure.diurnalBaseTime and ObservingProcedure.temporalObservingInterval (rename .samplingInterval to .temporalObservingInterval[0..1])
- Add placeholders for future Provenance sub-model based on OGC PROV-O building block. Add aggregationInterval as a note.
- re-insert ReportingProcedure.temporalReportingInterval[0..1]
- rename ReportingProcedure.spatialReportingInterval to .spatialReportingResolution
- add ObservingProcedure.spatialObservingResolution[0..1]

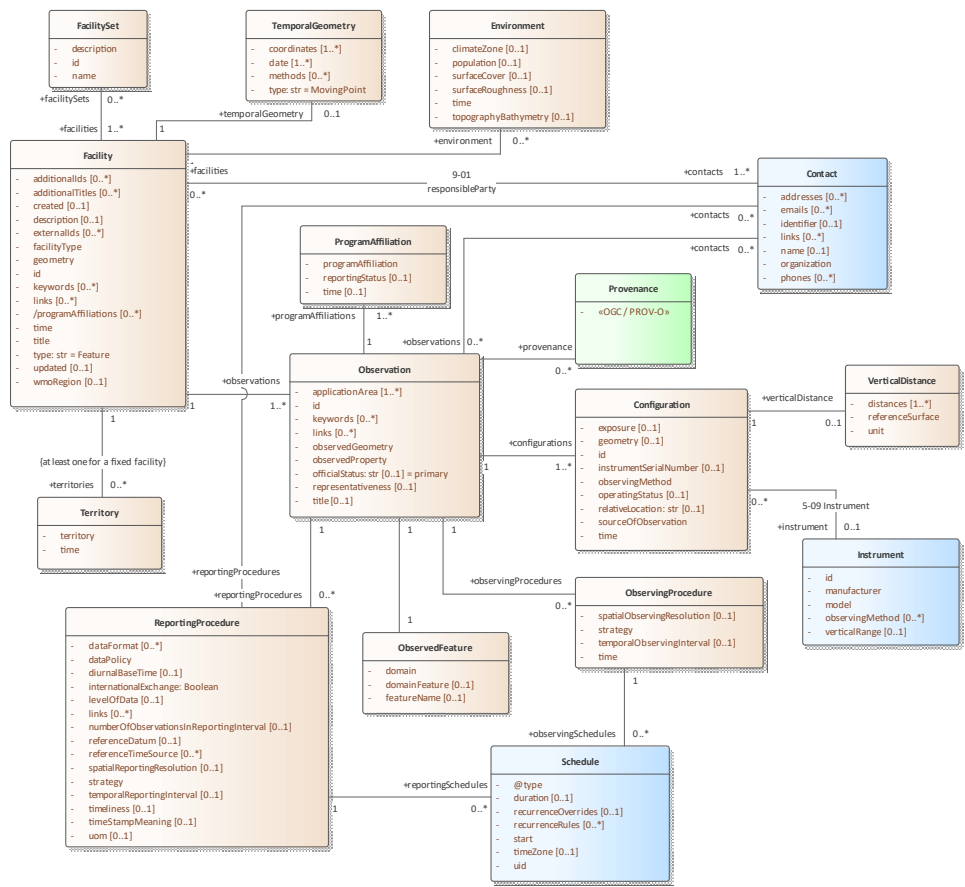


Figure 1: WMDR2

Configuration

Class in package 'WMDR2'

WMDR2 core JSON propoerty: configurations.
Describes the historical configuration of the observations during a specified time period.

Configuration

Version 1.0 Phase 1.0 Proposed

jkl created on 26.06.2026. Last modified 21.09.2026

ATTRIBUTES	
	exposure : Private

ATTRIBUTES	
Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False) Values from http://codes.wmo.int/wmdr/Exposure [Is static True. Containment is Not Specified.]	
 geometry : Private Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False) geometry is the OGC Records name for geolocation. For stationary facilities, this represents the geolocation of the sensor / inlet, and should be documented. If not specified, the geometry (and temporalGeometry) of the facility is assumed. For mobile facilities, this should be left blank. [Is static True. Containment is Not Specified.]	
 id : Private [Is static True. Containment is Not Specified.]	
 instrumentSerialNumber : Private Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False) [Is static True. Containment is Not Specified.]	
 observingMethod : Private from (new) unified code list of observing methods, based on http://codes.wmo.int/wmdr/ObservingMethodAtmosphere http://codes.wmo.int/wmdr/ObservingMethodTerrestrial http://codes.wmo.int/wmdr/ObservingMethodEarth [Is static True. Containment is Not Specified.]	
 operatingStatus : Private Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False) values refer to http://codes.wmo.int/wmdr/InstrumentOperatingStatus [Is static True. Containment is Not Specified.]	
 relativeLocation : str Private Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False) Free text, documenting the relative location of instrument / sensor / inlet /observer with respect to the geometry (and temporalGeometry) of the facility. [Is static True. Containment is Not Specified.]	
 sourceOfObservation : Private Value from http://codes.wmo.int/wmdr/SourceOfObservation. If sourceOfObservation=="humanObservation", several attributes of "observations" are expected to be null. [Is static True. Containment is Not Specified.]	
 time : Private conformant to OGC time: [beginDate, endDate] [Is static True. Containment is Not Specified.]	

ATTRIBUTES	
ASSOCIATIONS	
Association (direction: Unspecified) 5-09 Instrument Source: Public (Class) Configuration Cardinality: [0..*] Target: Public instrument (Class) Instrument Cardinality: [0..1]	
Association (direction: Unspecified) Source: Public (Class) Configuration Cardinality: [1] Target: Public verticalDistance (Class) VerticalDistance Cardinality: [0..1]	
Association (direction: Unspecified) Source: Public (Class) Observation Cardinality: [1] Target: Public configurations (Class) Configuration Cardinality: [1..*]	

Contact

Class in package 'WMDR2'

WMDR2 core JSON class: contact.

Contact associations carry one or more contextual **roles**. In JSON, the relationship is serialized as a `contactAssignments` object containing the contact reference and role(s).

Contact

Version 1.0 Phase 1.0 Proposed

jkl created on 11.03.2026. Last modified 25.09.2026

ATTRIBUTES	
addresses : Private Multiplicity: ([0..*], Allow duplicates: 0, Is ordered: False) Postal addresses. [Is static True. Containment is Not Specified.]	
emails : Private Multiplicity: ([0..*], Allow duplicates: 0, Is ordered: False) Email contacts. [Is static True. Containment is Not Specified.]	
identifier : Private Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False) Optional stable contact identifier. [Is static True. Containment is Not Specified.]	

ATTRIBUTES	
 links : Private Multiplicity: ([0..*], Allow duplicates: 0, Is ordered: False) Online resources associated with the contact. [Is static True. Containment is Not Specified.]	
 name : Private Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False) Either a full name, or a sequence of name parts. Examples: Dr. Jörg Klausen (this corresponds to FN in vcard) Klausen;Jörg;Dr.; (this corresponds to N in vcard) [Is static True. Containment is Not Specified.]	
 organization : Private Example: ORG:Federal Office of Meteorology and Climatology MeteoSwiss [Is static True. Containment is Not Specified.]	
 phones : Private Multiplicity: ([0..*], Allow duplicates: 0, Is ordered: False) Phone contacts. [Is static True. Containment is Not Specified.]	

ASSOCIATIONS	
 Association (direction: Unspecified) 9-01 responsibleParty Source: Public facilities (Class) Facility Cardinality: [0..*] Target: Public contacts (Class) Contact Cardinality: [1..*]	
 Association (direction: Unspecified) Source: Public observations (Class) Observation Cardinality: [0..*] Target: Public contacts (Class) Contact Cardinality: [0..*]	
 Association (direction: Unspecified) Source: Public reportingProcedures (Class) ReportingProcedure Cardinality: [0..*] Target: Public contacts (Class) Contact Cardinality: [0..*]	

Environment

Class in package 'WMDR2'

Facility-environment object containing climate, surface-cover, population, and topography/bathymetry histories.

Environment
Version 1.0 Phase 1.0 Proposed
ChatGPT created on 02.06.2026. Last modified 25.06.2026

ATTRIBUTES	
 climateZone : Private Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False)	[Is static True. Containment is Not Specified.]
 population : Private Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False) Population in 10 and 50 km radius around the facility	[Is static True. Containment is Not Specified.]
 surfaceCover : Private Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False)	[Is static True. Containment is Not Specified.]
 surfaceRoughness : Private Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False)	[Is static True. Containment is Not Specified.]
 time : Private conformant to OGC time: [beginDate, endDate]	[Is static True. Containment is Not Specified.]
 topographyBathymetry : Private Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False)	[Is static True. Containment is Not Specified.]
ASSOCIATIONS	
 Association (direction: Unspecified) Source: Public (Class) Facility Cardinality: [1]	Target: Public environment (Class) Environment Cardinality: [0..*]

Facility

Class in package 'WMDR2'

Facility properties payload of the WMDR2 full record. JSON serialization is split between GeoJSON root and properties.

Facility
Version 1.0 Phase 1.0 Proposed
jkl created on 06.03.2026. Last modified 25.09.2026

ATTRIBUTES	
 additionalIds : Private Multiplicity: ([0..*], Allow duplicates: 0, Is ordered: False)	
Additional WSIs for the facility.	[Is static True. Containment is Not Specified.]
 additionalTitles : Private Multiplicity: ([0..*], Allow duplicates: 0, Is ordered: False)	
Additional names for the facility, i.e., "aliases" that may be in use, maybe by specific programs.	[Is static True. Containment is Not Specified.]
 created : Private Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False)	
Record creation date/time metadata.	[Is static True. Containment is Not Specified.]
 description : Private Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False)	
Facility description.	[Is static True. Containment is Not Specified.]
 externalIds : Private Multiplicity: ([0..*], Allow duplicates: 0, Is ordered: False)	
Generic OGC Records metadata. External identifiers such as used internally by an organization's data management system; also used to document programSpecificFacilityId. In that case, the "scheme" should be a recognized program, e.g., "externalIds": [{ "scheme": "https://codes.wmo.int/wmdr/ProgramAffiliation/GAWglobal", "value": "JFJ" }, { "scheme": "https://codes.wmo.int/wmdr/ProgramAffiliation/NDACC", "value": "Jungfrauoch, Switzerland" }]	
Note: Any programSpecificFacilityTitle should be listed under additionalTitles.	[Is static True. Containment is Not Specified.]
 facilityType : Private Alias: 3-04 facilityType	
WMDR facility type code value.	[Is static True. Containment is Not Specified.]
 geometry : Private Alias: 3-07 current geolocation	

ATTRIBUTES	
geometry = geolocation = coordinates + elevation [Is static True. Containment is Not Specified.]	
💎 id : Private Alias: 3-06 (primary) WSI The primary WIGOS Station Identifier [Is static True. Containment is Not Specified.]	
💎 keywords : Private Multiplicity: ([0..*], Allow duplicates: 0, Is ordered: False) Discovery/search keywords retained in the core JSON. [Is static True. Containment is Not Specified.]	
💎 links : Private Multiplicity: ([0..*], Allow duplicates: 0, Is ordered: False) Related links. [Is static True. Containment is Not Specified.]	
💎 programAffiliations : Private Multiplicity: ([0..*], Allow duplicates: 0, Is ordered: False) Facility programme affiliation and programme-specific reporting status are derived from the programme affiliations and reporting statuses of the Facility's Observations according to WMDR business rules. [Is static True. Containment is Not Specified.]	
💎 time : Private Alias: [dateEstablished, dateClosed] mandatory element in OGC Records API Part 1, an interval. Used to declare [dateEstablished, dateClosed] [Is static True. Containment is Not Specified.]	
💎 title : Private Alias: 3-03 facilityName properties.title = 3-03 facilityName [Is static True. Containment is Not Specified.]	
💎 type : str Private = Feature Facility properties type; const facility. [Is static True. Containment is Not Specified.]	
💎 updated : Private Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False) Record update date/time metadata. [Is static True. Containment is Not Specified.]	

ATTRIBUTES	
wmoRegion : Private Alias: 3-01 wmoRegion Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False) mandatory for fixed facilities, optional for mobile facilities. [Is static True. Containment is Not Specified.]	
ASSOCIATIONS	
Association (direction: Unspecified) 9-01 responsibleParty Source: Public facilities (Class) Facility Cardinality: [0..*] Target: Public contacts (Class) Contact Cardinality: [1..*]	
Association (direction: Unspecified) Source: Public (Class) Facility Cardinality: [1] Target: Public observations (Class) Observation Cardinality: [1..*]	
Association (direction: Unspecified) Source: Public facilities (Class) Facility Cardinality: [1..*] Target: Public facilitySets (Class) FacilitySet Cardinality: [0..*]	
Association (direction: Unspecified) Source: Public (Class) Facility Cardinality: [1] Target: Public environment (Class) Environment Cardinality: [0..*]	
Association (direction: Unspecified) Source: Public (Class) Facility Cardinality: [1] Target: Public temporalGeometry (Class) TemporalGeometry Cardinality: [0..1]	
Association (direction: Unspecified) Constraints: at least one for a fixed facility : Invariant Source: Public (Class) Facility Cardinality: [1] Target: Public territories (Class) Territory Cardinality: [0..*]	

FacilitySet

Class in package 'WMDR2'

Retained from previous UML model; not part of the current WMDR2 core full-record JSON schema.

jkl created on 06.03.2026. Last modified 25.06.2026

ATTRIBUTES	
 description : Private	[Is static True. Containment is Not Specified.]
 id : Private	[Is static True. Containment is Not Specified.]
 name : Private Alias: title	[Is static True. Containment is Not Specified.]
ASSOCIATIONS	
 Association (direction: Unspecified)	
Source: Public facilities (Class) Facility Cardinality: [1..*]	Target: Public facilitySets (Class) FacilitySet Cardinality: [0..*]

Instrument

Class in package 'WMDR2'

Describes instrument capabilities, not a specific instrument instance.
Instrument are re-usable between configurations and even facilities.

Instrument
Version 1.0 Phase 1.0 Proposed
jkl created on 22.05.2026. Last modified 21.09.2026

ATTRIBUTES	
 id : Private	[Is static True. Containment is Not Specified.]
 manufacturer : Private Alias: 5-09 manufacturer	[Is static True. Containment is Not Specified.]
 model : Private Alias: 5-09 model	[Is static True. Containment is Not Specified.]
 observingMethod : Private Multiplicity: ([0..*], Allow duplicates: 0, Is ordered: False) Observing method(s) an instrument may expose. This is meant to support users in selecting the right instrument from a catalogue.	[Is static True. Containment is Not Specified.]

ATTRIBUTES	
verticalRange : Private Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False) Specify using an array of [lower boundary, upper boundary] in meter. This is a capability of an instrument as specified, not the actually accessible vertical range. [Is static True. Containment is Not Specified.]	
ASSOCIATIONS	
Association (direction: Unspecified) 5-09 Instrument Source: Public (Class) Configuration Cardinality: [0..*] Target: Public instrument (Class) Instrument Cardinality: [0..1]	

Observation

Class in package 'WMDR2'

WMDR2 core JSON class: observations.
Describes a series of observations of a specific observedProperty with its observedGeometry.

Observation

Version 1.0 Phase 1.0 Proposed

jkl created on 06.03.2026. Last modified 21.09.2026

ATTRIBUTES	
applicationArea : Private Multiplicity: ([1..*], Allow duplicates: 0, Is ordered: False) Value(s) from http://codes.wmo.int/wmdr/ApplicationArea [Is static True. Containment is Not Specified.]	
id : Private Observation identifier, e.g. "12006-point". [Is static True. Containment is Not Specified.]	
keywords : Private Multiplicity: ([0..*], Allow duplicates: 0, Is ordered: False) Observation keywords. [Is static True. Containment is Not Specified.]	
links : Private Multiplicity: ([0..*], Allow duplicates: 0, Is ordered: False) Observation links. [Is static True. Containment is Not Specified.]	

ATTRIBUTES	
 observedGeometry : Private cf. http://codes.wmo.int/wmdr/Geometry	[Is static True. Containment is Not Specified.]
 observedProperty : Private Alias: OMS:ObservedProperty, iop:Variable The observedVariable is constructed following I-ADOPT ontology approaches, with elements described in a unified vocabulary hosted at codes.wmo.int/wmdr2 . NB: observedVariable will be a LONG list of entries, each record comprised of notation, name, description. The notation is a unique identifier, not necessarily human-readable. The name is potentially human-readable. The description is a fully fledged definition, human-readable and describing all relevant aspects of an observed variable. It is derived from the observedProperty, the object of interest, the matrix and domain, as well as constraints. Matrix and constraints are defined in controlled vocabularies that may be domain-specific. Examples include: - observableProperty: state variable (e.g., temperature, pressure, leaf-area-index (LAI), ...), or another property (e.g., thickness, concentration, scattering coefficient, color, ...) - objectOfInterest: sea ice, CO2, aerosol particles, chlorophyll - observablePropertyConstraints: a conventional level above ground (e.g., 2m above ground for temperature measurements), a size fraction (e.g., PM2.5), a specific wavelength (e.g., 635 nm), a pressure (range), a temperature (range), a humidity (range), etc., as relevant for the result of the observation. Constraints shall be documented as name-value pairs, where the names shall come from a code list, and the values shall come from a code list depending on the name, or be free text. - matrix: atmosphere, interstitial air in the root space, air in an ice core, sea surface, soil, dry air, ... These elements will be supported by code lists from codes.wmo.int/wmdr2 . Applications may want to guide users to select a specific observedVariable by offering groupings, filtering, or tree-like visualizations according to matrix, objectOfInterest, etc.	[Is static True. Containment is Not Specified.]
 officialStatus : str Private = primary Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False) Values from http://codes.wmo.int/wmdr/ObservationStatus Defaults to "primary" if not provided.	[Is static True. Containment is Not Specified.]
 representativeness : Private Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False) http://codes.wmo.int/wmdr/Representativeness	[Is static True. Containment is Not Specified.]
 title : Private Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False) Human-readable title derived from domain, geometry, observed variable, and optional label.	[Is static True. Containment is Not Specified.]

ASSOCIATIONS	
 Association (direction: Unspecified)	
Source: Public (Class) Observation	Target: Public observingProcedures (Class)

ASSOCIATIONS	
Cardinality: [1]	ObservingProcedure Cardinality: [0..*]
 Association (direction: Unspecified)	
Source: Public (Class) Observation Cardinality: [1]	Target: Public programAffiliations (Class) ProgramAffiliation Cardinality: [1..*]
 Association (direction: Unspecified)	
Source: Public (Class) Observation Cardinality: [1]	Target: Public configurations (Class) Configuration Cardinality: [1..*]
 Association (direction: Unspecified)	
Source: Public observations (Class) Observation Cardinality: [0..*]	Target: Public contacts (Class) Contact Cardinality: [0..*]
 Association (direction: Unspecified)	
Source: Public (Class) Observation Cardinality: [1]	Target: Public (Class) ObservedFeature Cardinality: [1]
 Association (direction: Unspecified)	
Source: Public (Class) Observation	Target: Public provenance (Class) Provenance Cardinality: [0..*]
 Association (direction: Unspecified)	
Source: Public (Class) Observation Cardinality: [1]	Target: Public reportingProcedures (Class) ReportingProcedure Cardinality: [0..*]
 Association (direction: Unspecified)	
Source: Public (Class) Facility Cardinality: [1]	Target: Public observations (Class) Observation Cardinality: [1..*]

ObservedFeature

Class in package 'WMDR2'

Describes the physical domain, i.e. larger-scale environmental target or ultimate-feature-of-interest, of an observation.

ObservedFeature

Version 1.0 Phase 1.0 Proposed

jkl created on 10.06.2026. Last modified 25.06.2026

ATTRIBUTES	
 domain : Private	
The largest scale in WIGOS, cf. http://codes.wmo.int/wmdr/Domain	[Is static True. Containment is Not Specified.]
 domainFeature : Private Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False)	
Describes the type of the larger-scale environmental target or ultimate-feature-of-interest, of an observation.	
Concept motivated by requirements from hydrology, but may be useful elsewhere. New code list is proposed, with terms like	
- river basin	
- lake	
- river	
- catchment	
- canopy	
- epilimnion	
- hypolimnion	
- troposphere	
- tropopause	
- stratosphere	
- thermocline (useful as a general term, but needs "domain")	
- boundary layer (useful as a general term, but needs "domain")	
- ...	[Is static True. Containment is Not Specified.]
 featureName : Private Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False)	
free text field, e.g., "Danube Basin", "Amazon rain forest", "Amazon river Basin"	[Is static True. Containment is Not Specified.]
ASSOCIATIONS	
 Association (direction: Unspecified)	
Source: Public (Class) Observation Cardinality: [1]	Target: Public (Class) ObservedFeature Cardinality: [1]

ObservingProcedure

Class in package 'WMDR2'

When are observations made? Describes an observingStrategy, and depending on it, a temporal schedule/coverage, e.g. "from Feb-Oct on working days from 06-18 UTC".

Implement as array of objects

ObservingProcedure
Version 1.0 Phase 1.0 Proposed

CONSTRAINTS	
 Invariant. {temporalObservingInterval is specified OR at least one observingSchedule is specified} [Approved, Weight is 0.]	
ATTRIBUTES	
 spatialObservingResolution : Private Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False) To be implemented as {"value": [float], "uom": str}, whereby len([float])<3 to report either a distance or a grid / pixel -dimension. [Is static True. Containment is Not Specified.]	
 strategy : Private Values from http://codes.wmo.int/wmdr/SamplingStrategy [Is static True. Containment is Not Specified.]	
 temporalObservingInterval : Private Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False) ISO-8601 duration, e.g. - "PT10M" for 10' intervals - "PT1H" for 1-hour intervals - "P1D" for daily intervals [Is static True. Containment is Not Specified.]	
 time : Private conformant to OGC time: [beginDate, endDate] [Is static True. Containment is Not Specified.]	
ASSOCIATIONS	
 Association (direction: Unspecified) Source: Public (Class) ObservingProcedure Cardinality: [1] Target: Public observingSchedules (Class) Schedule Cardinality: [0..*]	
 Association (direction: Unspecified) Source: Public (Class) Observation Cardinality: [1] Target: Public observingProcedures (Class) ObservingProcedure Cardinality: [0..*]	

ProgramAffiliation

Class in package 'WMDR2'

ProgramAffiliation
Version 1.0 Phase 1.0 Proposed
ChatGPT created on 01.06.2026. Last modified 25.06.2026
Alias temporalProgramAffiliation

ATTRIBUTES	
 programAffiliation : Private	[Is static True. Containment is Not Specified.]
 reportingStatus : Private Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False) http://codes.wmo.int/wmdr/ReportingStatus	[Is static True. Containment is Not Specified.]
 time : Private Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False)	[Is static True. Containment is Not Specified.]
ASSOCIATIONS	
 Association (direction: Unspecified) Source: Public (Class) Observation Cardinality: [1]	Target: Public programAffiliations (Class) ProgramAffiliation Cardinality: [1..*]

Provenance

Class in package 'WMDR2'

[TODO] Provenance is supplied by the OGC provenance building block

- aggregationInterval [0..1]: Describes the smallest interval over which raw/ level 0 observations are aggregated and stored. Express as ISO-8601 duration.

Provenance
Version 1.0 Phase 1.0 Proposed
jkl created on 24.09.2026. Last modified 25.09.2026

ATTRIBUTES	
 «OGC / PROV-O»: Private	[Is static True. Containment is Not Specified.]
ASSOCIATIONS	
 Association (direction: Unspecified) Source: Public (Class) Observation	Target: Public provenance (Class) Provenance Cardinality: [0..*]

ASSOCIATIONS

ReportingProcedure

Class in package 'WMDR2'

ReportingProcedure

Version 1.0 Phase 1.0 Proposed

jkl created on 23.06.2026. Last modified 25.09.2026

CONSTRAINTS
 Invariant. {temporalReportingInterval is specified OR at least one reportingSchedule is specified} [Approved, Weight is 0.]

ATTRIBUTES
 dataFormat : Private Multiplicity: ([0..*], Allow duplicates: 0, Is ordered: False) Aligned array, values from http://codes.wmo.int/wmdr/DataFormat. [Is static True. Containment is Not Specified.]
 dataPolicy : Private Value from http://codes.wmo.int/wmdr/DataPolicy. JSON Concept or explicit null where permitted. [Is static True. Containment is Not Specified.]
 diurnalBaseTime : Private Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False) [Is static True. Containment is Not Specified.]
 internationalExchange : Boolean Private Declaration if data are intended for international exchange or not. [Is static True. Containment is Not Specified.]
 levelOfData : Private Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False) Value from http://codes.wmo.int/wmdr/LevelOfData. [Is static True. Containment is Not Specified.]
 links : Private Multiplicity: ([0..*], Allow duplicates: 0, Is ordered: False) URL to related WIS record, entry point for data discovery and retrieval. [Is static True. Containment is Not Specified.]

ATTRIBUTES	
🔹 numberOfObservationsInReportingInterval : Private Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False)	[Is static True. Containment is Not Specified.]
🔹 referenceDatum : Private Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False) Value from http://codes.wmo.int/wmdr/CoordinateReferenceSystem.	[Is static True. Containment is Not Specified.]
🔹 referenceTimeSource : Private Multiplicity: ([0..*], Allow duplicates: 0, Is ordered: False) Single value or array, values from http://codes.wmo.int/wmdr/ReferenceTime.	[Is static True. Containment is Not Specified.]
🔹 spatialReportingResolution : Private Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False) To be implemented as {"value": [float], "uom": str}, whereby len([float])<3 to report either a distance or a grid / pixel -dimension.	[Is static True. Containment is Not Specified.]
🔹 strategy : Private	[Is static True. Containment is Not Specified.]
🔹 temporalReportingInterval : Private Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False) ISO-8601 duration, e.g. "PT1H"	[Is static True. Containment is Not Specified.]
🔹 timeliness : Private Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False) Expected timeliness of reporting, an object with value and uom (default: seconds)	[Is static True. Containment is Not Specified.]
🔹 timeStampMeaning : Private Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False) Aligned array, values from http://codes.wmo.int/wmdr/TimeStampMeaning.	[Is static True. Containment is Not Specified.]
🔹 uom : Private Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False) Unit of reported values of the observedProperty. Values from http://codes.wmo.int/wmdr/unit.	[Is static True. Containment is Not Specified.]

ASSOCIATIONS	
 Association (direction: Unspecified)	
Source: Public (Class) ReportingProcedure Cardinality: [1]	Target: Public reportingSchedules (Class) Schedule Cardinality: [0..*]
 Association (direction: Unspecified)	
Source: Public reportingProcedures (Class) ReportingProcedure Cardinality: [0..*]	Target: Public contacts (Class) Contact Cardinality: [0..*]
 Association (direction: Unspecified)	
Source: Public (Class) Observation Cardinality: [1]	Target: Public reportingProcedures (Class) ReportingProcedure Cardinality: [0..*]

Schedule

Class in package 'WMDR2'

Reusable JSCalendar Event schedule to document temporal coverage and exceptions.

Schedule

Version 1.0 Phase 1.0 Proposed

ChatGPT created on 25.06.2026. Last modified 25.09.2026

ATTRIBUTES	
 @type : Private	
JSCalendar object type; const Event.	[Is static True. Containment is Not Specified.]
 duration : Private	
Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False)	
Occurrence duration describing coverage window or sampling duration.	[Is static True. Containment is Not Specified.]
 recurrenceOverrides : Private	
Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False)	
JSCalendar recurrence override patches; null override values are invalid.	[Is static True. Containment is Not Specified.]
 recurrenceRules : Private	
Multiplicity: ([0..*], Allow duplicates: 0, Is ordered: False)	
JSCalendar recurrence rules. Use start, duration and other elements to evaluate constraints if needed.	[Is static True. Containment is Not Specified.]

ATTRIBUTES	
start : Private Canonical JSCalendar recurrence anchor; date 0001-01-01 with time from coverage start. [Is static True. Containment is Not Specified.]	
timeZone : Private Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False) IANA time zone or UTC default. [Is static True. Containment is Not Specified.]	
uid : Private JSCalendar-safe schedule uid. [Is static True. Containment is Not Specified.]	

ASSOCIATIONS	
Association (direction: Unspecified) Source: Public (Class) ObservingProcedure Cardinality: [1]	Target: Public observingSchedules (Class) Schedule Cardinality: [0..*]
Association (direction: Unspecified) Source: Public (Class) ReportingProcedure Cardinality: [1]	Target: Public reportingSchedules (Class) Schedule Cardinality: [0..*]

TemporalGeometry

Class in package 'WMDR2'

TemporalGeometry

Version 1.0 Phase 1.0 Proposed

jkl created on 11.03.2026. Last modified 25.09.2026

Alias temporalGeometry

ATTRIBUTES	
coordinates : Private Multiplicity: ([1..*], Allow duplicates: 0, Is ordered: False) Coordinate history array; each entry is lon, lat[, elevation]. [Is static True. Containment is Not Specified.]	
date : Private Multiplicity: ([1..*], Allow duplicates: 0, Is ordered: False)	

ATTRIBUTES	
Date array aligned with coordinates.	[Is static True. Containment is Not Specified.]
methods : Private Multiplicity: ([0..*], Allow duplicates: 0, Is ordered: False) use http://codes.wmo.int/wmdr/GeopositioningMethod	[Is static True. Containment is Not Specified.]
type : str Private = MovingPoint	[Is static True. Containment is Not Specified.]

ASSOCIATIONS	
Association (direction: Unspecified) Source: Public (Class) Facility Cardinality: [1] Target: Public temporalGeometry (Class) TemporalGeometry Cardinality: [0..1]	

Territory

Class in package 'WMDR2'

WMDS reference: 3-02
mandatory for fixed facilities, optional for mobile facilities.
JSON serializes the temporal extent as official dates.

Territory

Version 1.0 Phase 1.0 Proposed

jkl created on 23.06.2026. Last modified 25.09.2026

ATTRIBUTES	
territory : Private Alias: temporal Territory Values from http://codes.wmo.int/wmdr/TerritoryName	[Is static True. Containment is Not Specified.]
time : Private conformant to OGC time: [beginDate, endDate]	[Is static True. Containment is Not Specified.]

ASSOCIATIONS	
Association (direction: Unspecified) Constraints: at least one for a fixed facility : Invariant	

ASSOCIATIONS	
Source: Public (Class) Facility Cardinality: [1]	Target: Public territories (Class) Territory Cardinality: [0..*]

VerticalDistance

Class in package 'WMDR2'

VerticalDistance

Version 1.0 Phase 1.0 Proposed

jkl created on 26.06.2026. Last modified 21.09.2026

ATTRIBUTES	
 distances : Private Multiplicity: ([1..*], Allow duplicates: 0, Is ordered: False)	[Is static True. Containment is Not Specified.]
 referenceSurface : Private	[Is static True. Containment is Not Specified.]
 unit : Private	[Is static True. Containment is Not Specified.]

ASSOCIATIONS	
 Association (direction: Unspecified)	
Source: Public (Class) Configuration Cardinality: [1]	Target: Public verticalDistance (Class) VerticalDistance Cardinality: [0..1]